Experimental Setup

Experimental Setup I

This section details the dataset, schema, and software environment used to produce the results. The exemplar deliberately avoids manuscript token injection: concrete numbers are reproduced by running the analysis script, and the figures are regenerated from tested code, so nothing here can silently drift from a hardcoded value.

Dataset I

The analysis uses a shipped, deterministic CSV fixture (`data/measurements.csv`) generated once with a fixed NumPy seed. It contains 120 subject records with the following columns:

Column	Role	Type
`subject_id`	per-row identifier	string
`group`	categorical group (`alpha`, `beta`, `gamma`)	string
`height_cm`	numeric feature	float
`weight_kg`	numeric feature	float
`resting_hr_bpm`	numeric feature	float

Dataset II

These roles are declared once in `src/eda/dataset.py::DatasetSchema`, which the statistics, correlation, and figure functions consult. The generating process makes weight depend positively on height (a strong correlation) while resting heart rate is only weakly related, and a few numeric cells are left blank to exercise the missing-data path.

Analysis conditions I

The experiment overlay (`experiment_plan.yaml`) declares three conditions:

- **raw_dataset** (reference) — as loaded, with missing cells as `NaN`.
- **cleaned_dataset** (proposed) — rows with any missing numeric feature dropped via `clean_dataset()`, which reports the count removed.
- **normalized_dataset** (variant) — the cleaned dataset with numeric columns z-score standardized by `normalize_numeric()` for cross-feature comparison.

The primary descriptive lens is the pairwise Pearson correlation among the numeric features.

Computational environment I

- **Language**: Python (see root `pyproject.toml` for the supported range).
- **Core dependencies**: `numpy`, `pandas`, `matplotlib` (declared in `domain_profile.yaml::required_packages`).
- **Headless plotting**: the analysis script sets `MPLBACKEND=Agg` before importing matplotlib.

Pipeline ordering I

The typical analysis order is:

1. `scripts/eda_analysis.py` — loads and cleans the dataset, then writes `output/figures/*.png` and `output/data/summary_statistics.csv`, printing each output path for manifest collection.
2. PDF rendering reads `manuscript/*.md` and `config.yaml` so figure paths and prose match the analysis that just completed.

Relation to figures I

Figure ([@sec:results])	Figure-data preparer (`src/eda/figures.py`)	Primary inputs
Height histogram	`histogram_data()`	`height_cm` column, 10 bins
Rows per group	`group_count_data()`	group column
Correlation heatmap	`correlation_heatmap_data()`	all numeric features

Relation to figures II

This table is descriptive documentation only; it is not executed as code during the build.